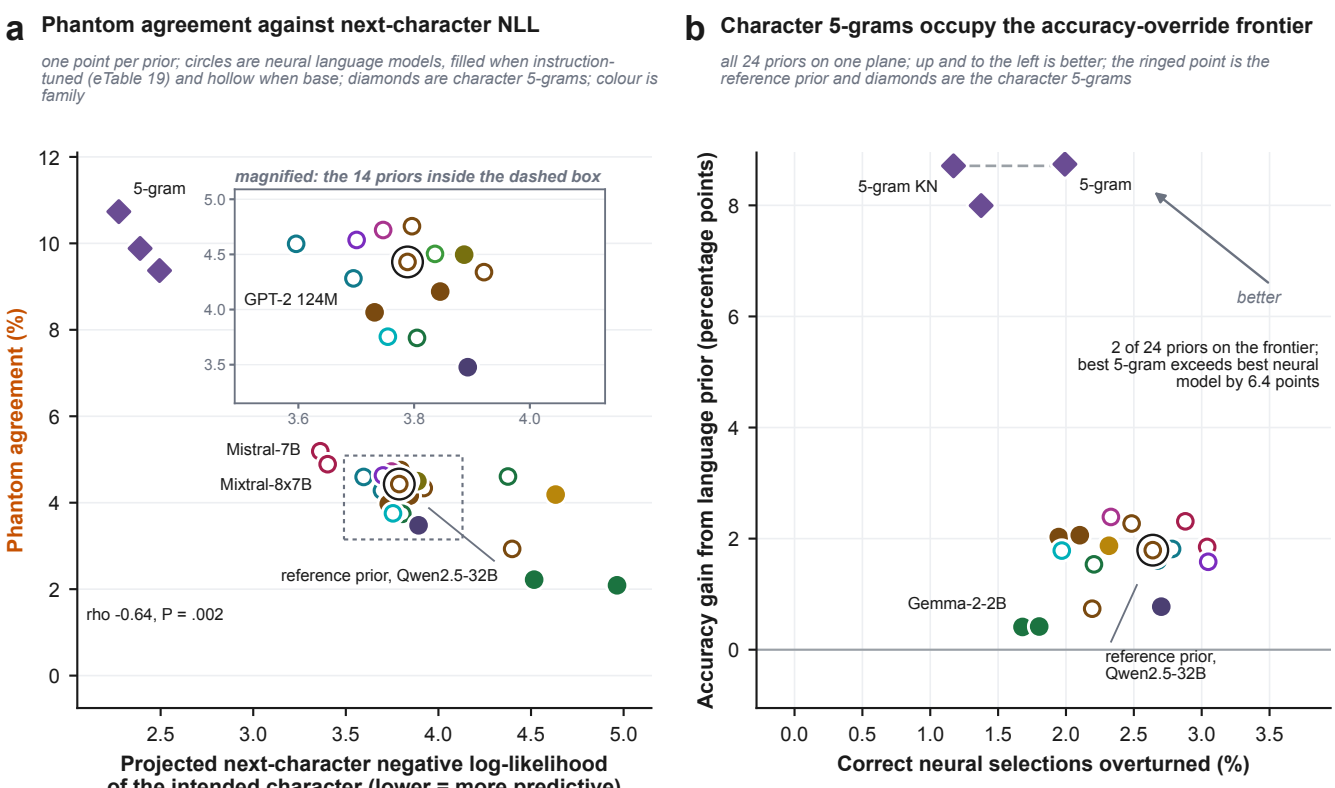

a
Phantom agreement against next-character NLL
one point per prior; circles are neural language models, filled when instruction-tuned (eTable 19) and hollow when base; diamonds are character 5-grams; colour is family
magnified: the 14 priors inside the dashed box
5-gram
GPT-2 124M
Mistral-7B
Mixtral-8x7B
reference prior, Qwen2.5-32B
rho -0.64, P = .002
Phantom agreement (%)
Projected next-character negative log-likelihood of the intended character (lower = more predictive)
b
Character 5-grams occupy the accuracy-override frontier
all 24 priors on one plane; up and to the left is better; the ringed point is the reference prior and diamonds are the character 5-grams
5-gram KN
5-gram
better
2 of 24 priors on the frontier; best 5-gram exceeds best neural model by 6.4 points
Gemma-2-2B
reference prior, Qwen2.5-32B
Accuracy gain from language prior (percentage points)
Correct neural selections overturned (%)